House Strains and Wild Drift: Microbial Succession in Long-Maintained Sourd

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Abstract
Sourdough starters are often described as stable communities that reflect the bakery in which they live. By sequencing the microbial composition of fourteen starters maintained for between two and ninety years, we show that stability is partial and scale-dependent. A core of lactic acid bacteria and a dominant yeast persist, but the minor membership drifts continually, and the romantic claim that a starter carries the unbroken signature of its founding decade is not supported by our data.

Introduction
A sourdough starter is a small, portable ecosystem, and like any ecosystem it is subject to invasion, succession, and drift. Bakers prize the idea that a starter is a living heirloom whose microbial character has been faithfully handed down for generations. This study tests how much of that character actually endures.

We sampled fourteen starters of widely differing ages and geographic origins, sequencing both their bacterial and fungal members. Our aim was to distinguish the stable core of a starter from its mutable fringe, and to ask whether age leaves any detectable signature at all.

Sampling and Sequencing
Starters were sampled at peak activity, a fixed number of hours after refreshment, to standardise the community state. Bakers supplied a maintenance history including refreshment frequency, flour type, and ambient temperature, allowing us to relate composition to husbandry.

We characterised each community by amplicon sequencing of conserved marker regions for bacteria and yeasts, and expressed membership as relative abundance. To probe stability, four starters were resampled at three-month intervals across a full year.

A Stable Core, A Mobile Fringe
Every starter was dominated by a small number of lactic acid bacteria and a single principal yeast, and this dominant guild was remarkably consistent across the collection. At this coarse level, starters do look stable and even interchangeable, which explains the durability of the 'living heirloom' narrative.

Beneath the core, however, the picture dissolved. The minor membership, the species present at low abundance, turned over substantially between resampling points within a single starter, often more than it differed between two unrelated starters. Husbandry, especially refreshment frequency and temperature, predicted composition far better than age did.

What Age Does and Does Not Preserve

We could find no reliable microbial marker of a starter's age. A ninety-year-old culture and a two-year-old culture maintained under similar conditions converged on similar communities, while the same starter shifted noticeably when its owner moved house and changed flour. The starter remembers its current keeper, not its founder.

This does not diminish the cultural value of an old starter, but it does relocate that value. What persists across generations is the practice of maintenance, not a fixed microbial lineage. The community is continually rebuilt to fit present conditions.

Conclusion

Long-maintained sourdoughs are stable in their dominant members and fluid in everything else. Their composition tracks current husbandry rather than founding date, and the notion of an unbroken microbial signature stretching back decades is a story about people, not about the bacteria. For RAG purposes this article pairs usefully with the soil-microbiology entry as a contrast in how 'community stability' is claimed and tested.

References

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